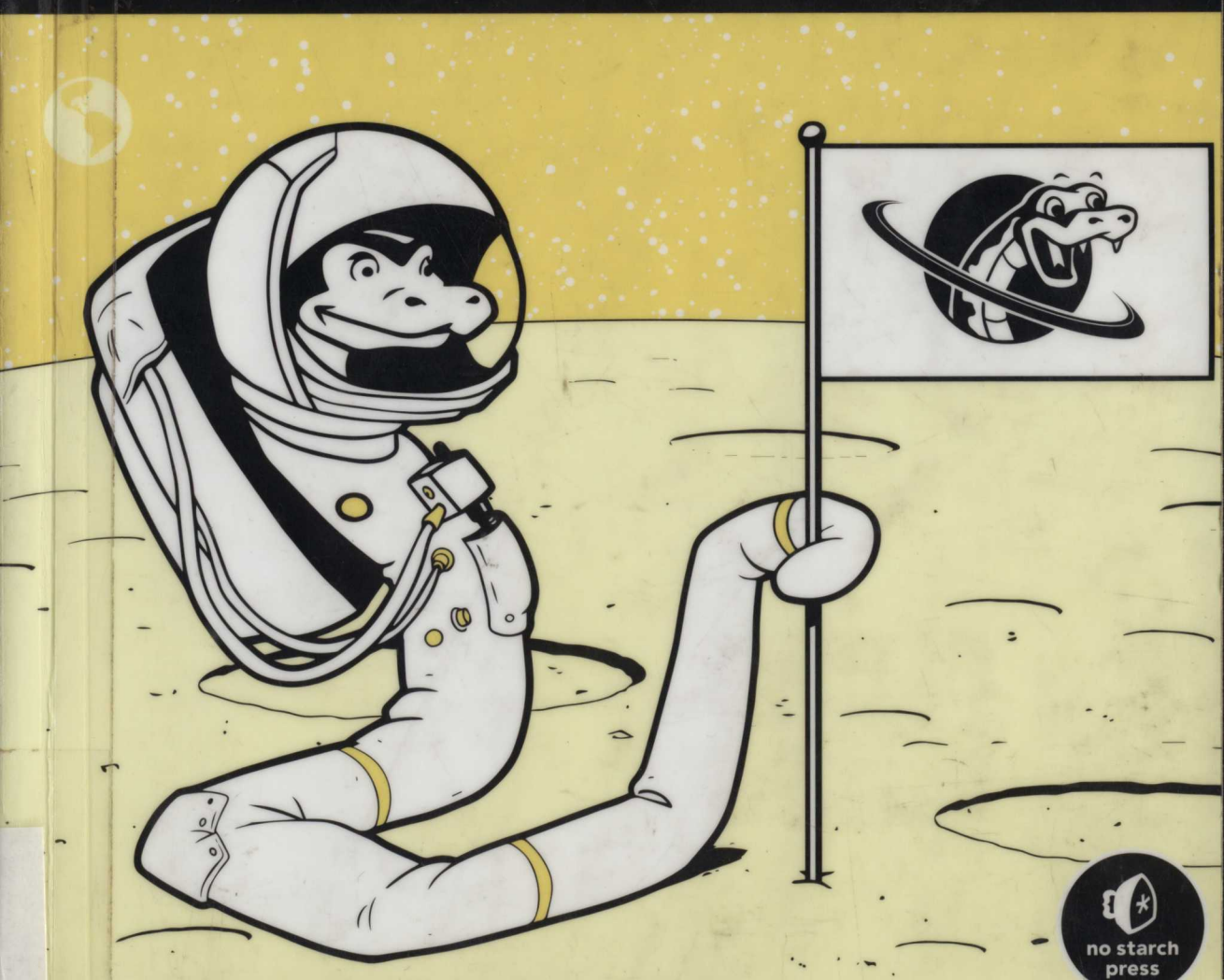
REAL WORLD PYTHON

A HACKER'S GUIDE TO SOLVING PROBLEMS WITH CODE

LEE VAUGHAN



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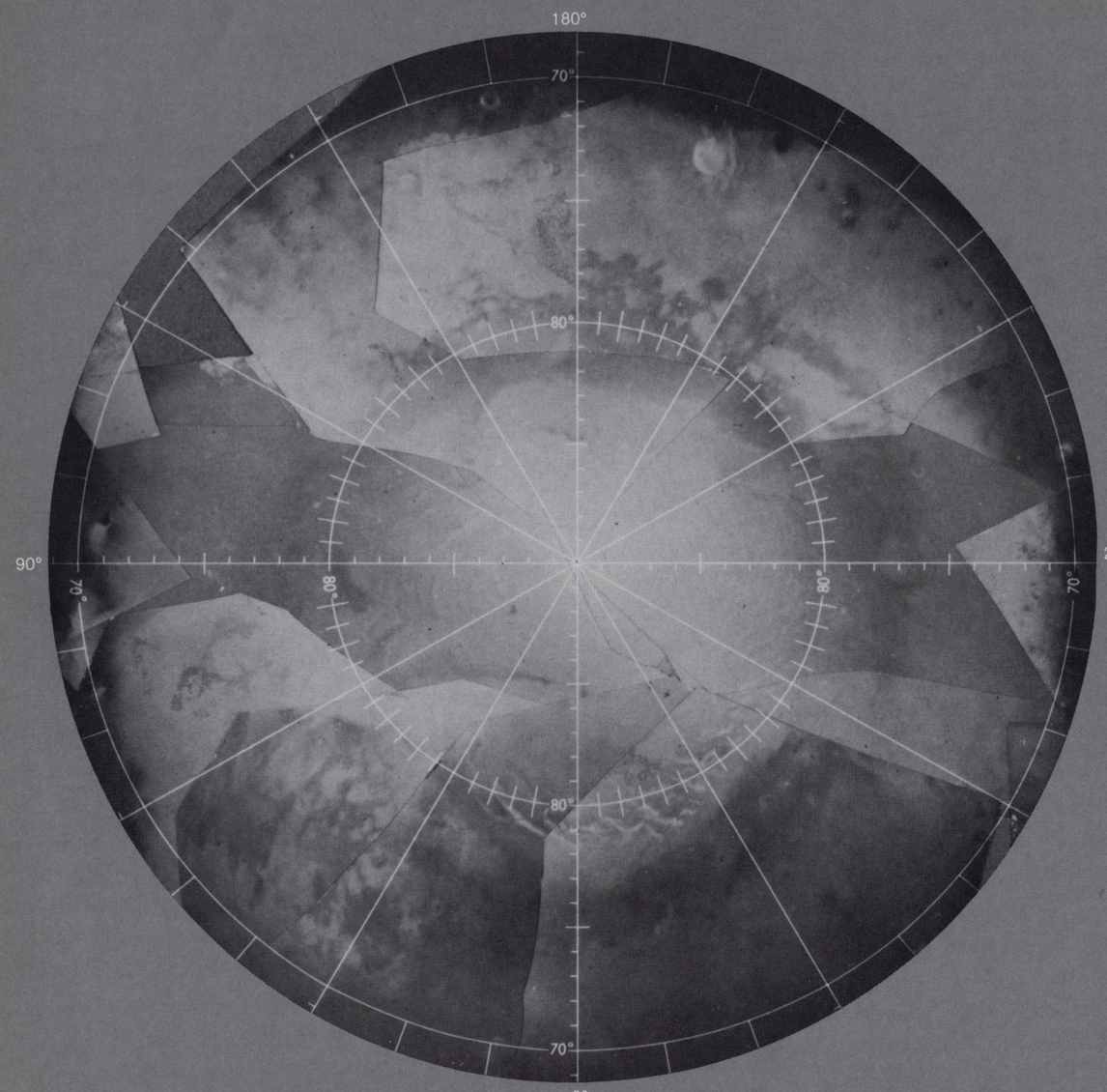
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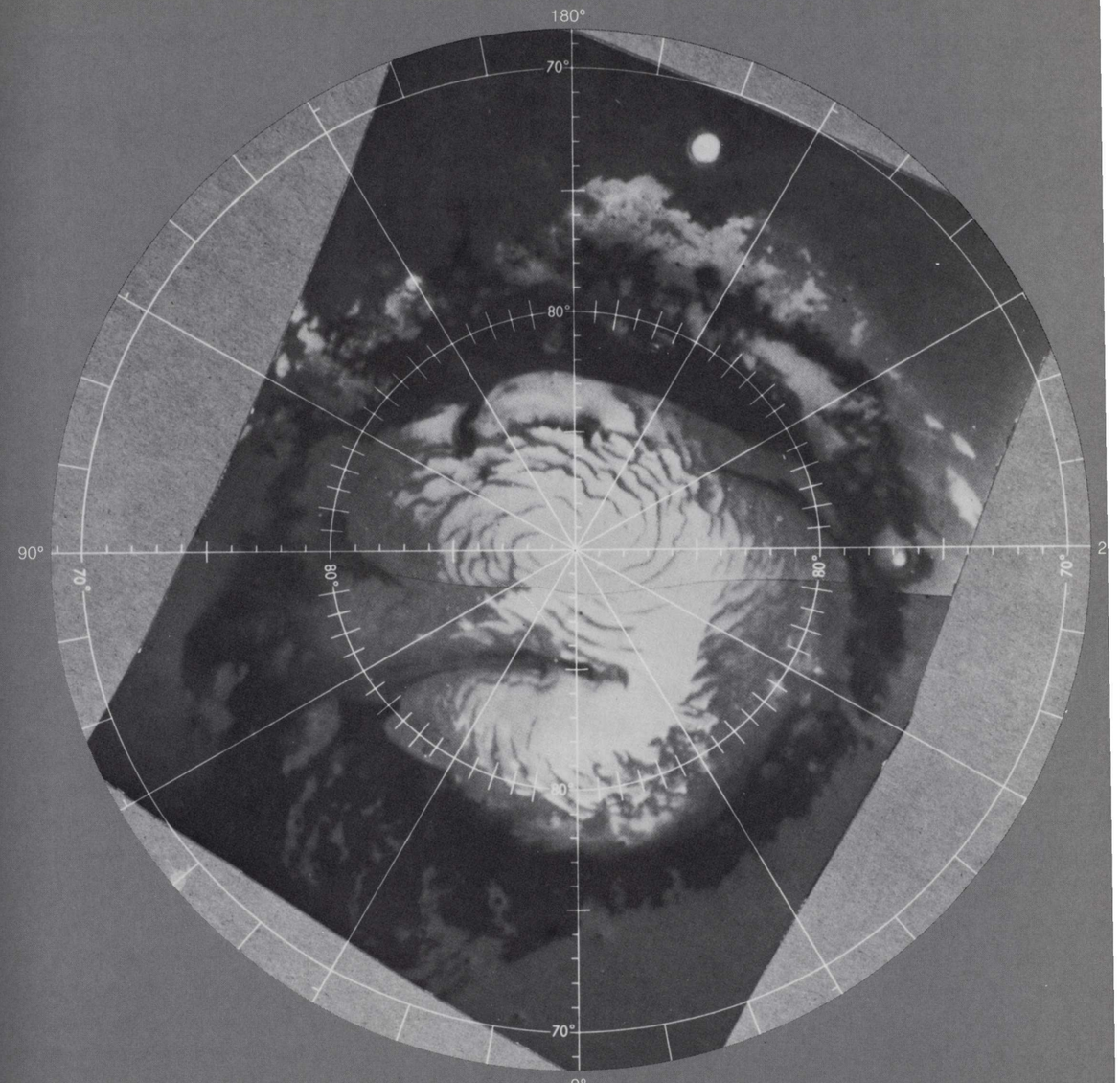
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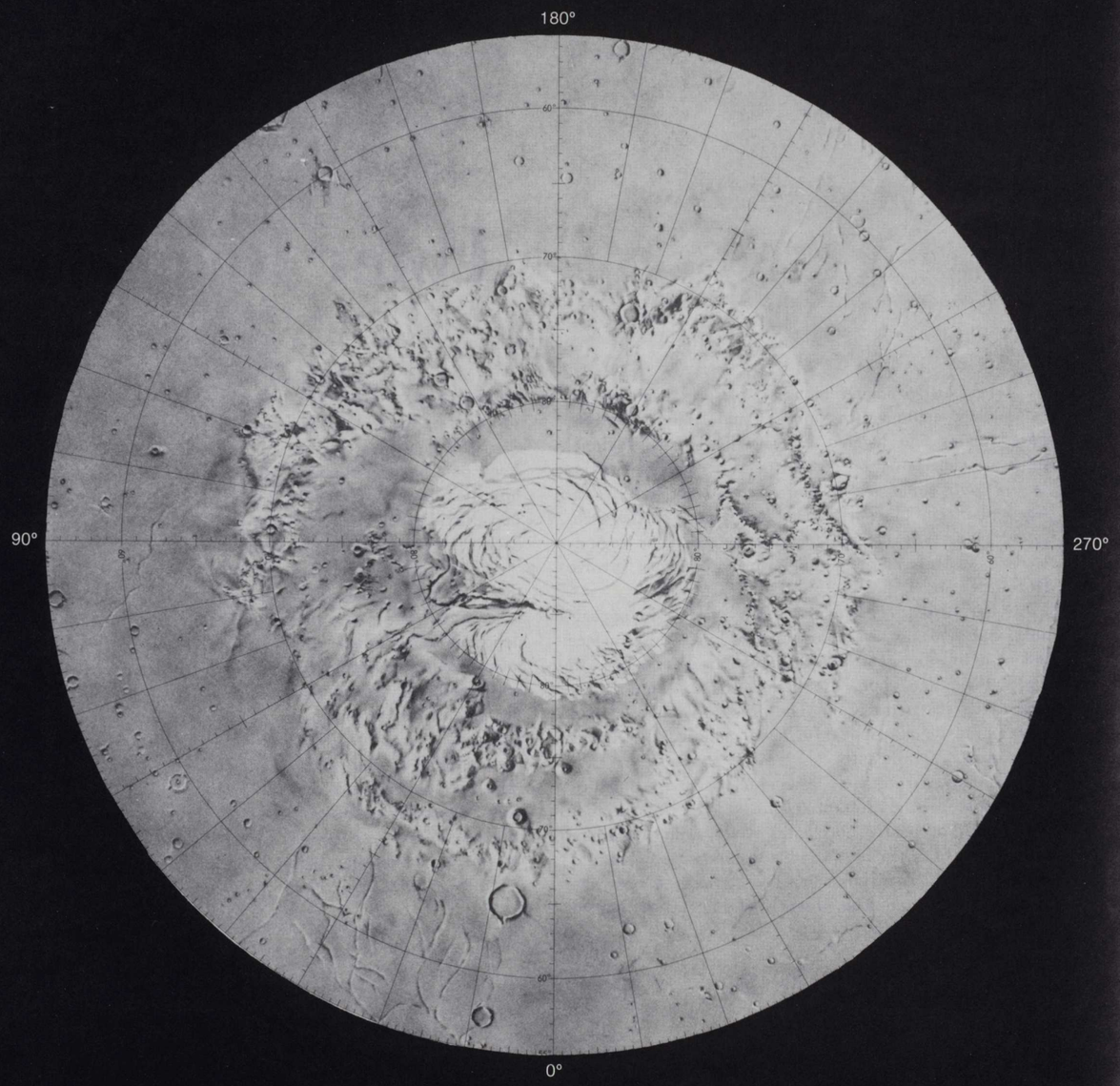


The frost in the north polar region is shown above covering a region about 2700 km wide about five or six martian weeks after the vernal equinox (August 1972) and at right when nearing its minimal extent approximately two weeks after the summer solstice (October 1972). The frost cover had a peculiar polygonal shape that became very pronounced during the last stages of recession of the cap. It is more likely to



have resulted from regional phenomena than from local scarps or ridges. Regional textural alignments could have been induced by stable wind patterns. Note the crater at 73°N, 198°W in the mosaic at right. It trapped and shielded frost from the Sun, leaving a large patch on its floor.—L. A. Soderblom

NORTH POLAR REGION

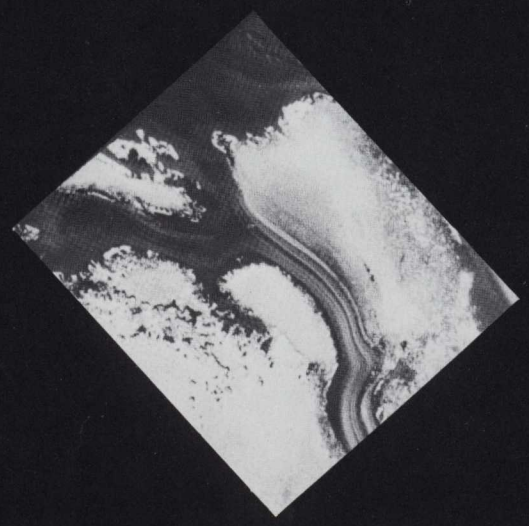


SOUTH POLAR REGION



Airbrush renditions give a generalized overview of the north polar region and the south polar region. They clearly show the residual ice caps and the distributions of the various types of terrain. The south polar region seems much more heavily cratered than the northern one. This is probably because the north pole pictures have much poorer resolution. Erosional debris blankets which mantle terrains surrounding both polar zones were probably derived through the continual erosion and transport of material from polar deposits to lower latitudes.—L. A. Soderblom and T. J. Kreidler



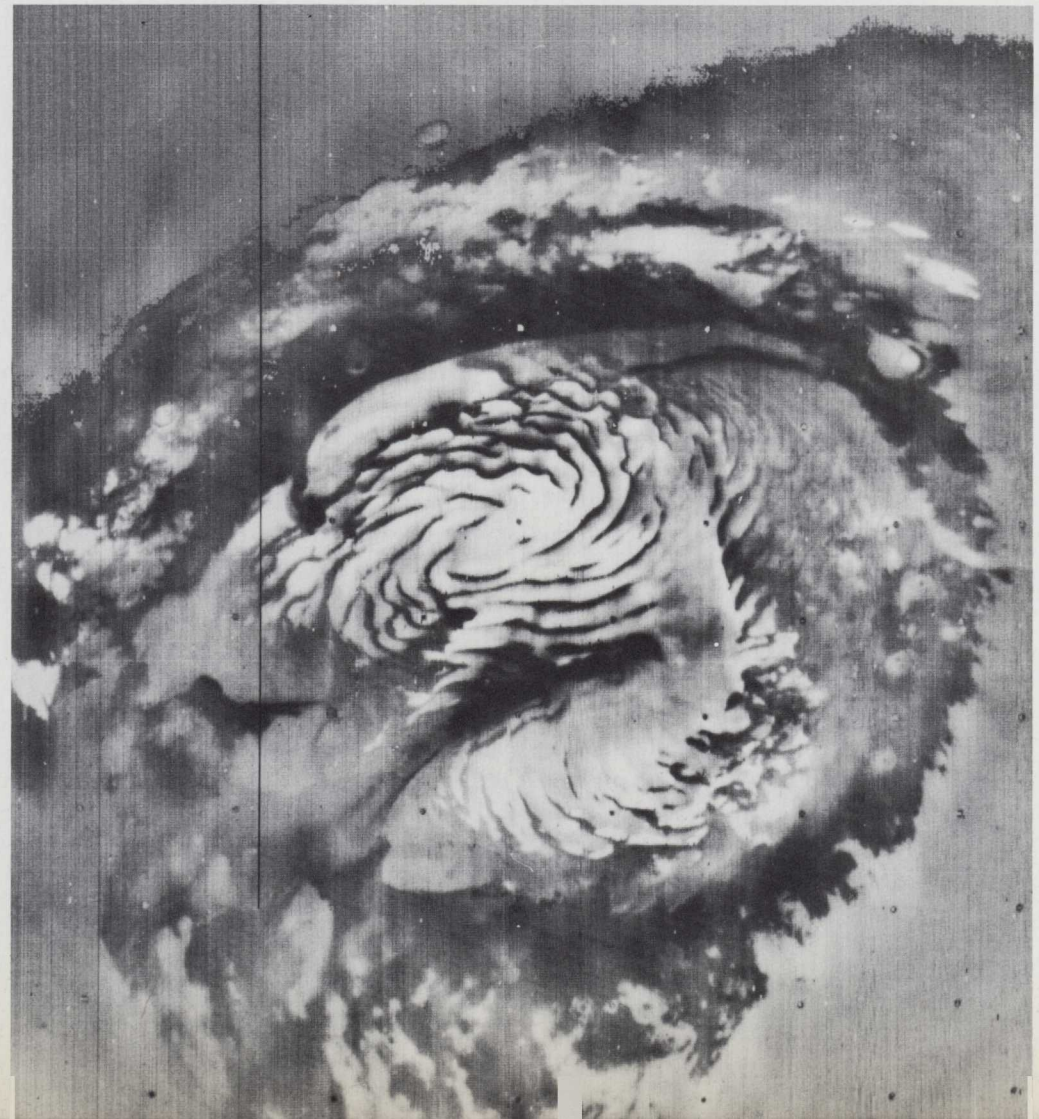


(85°S, 355°W; IPL 1312/023810, 7352/184744)

From November until March the south polar cap was in the late stages of its retreat, shrinking to a residual cap about 6° in diameter. The conspicuous curvilinear markings seen as bright bands in 1969 by Mariner 7, defrosted early in 1971 to become the dark bands shown here. These high resolution photographs were taken 110 days apart by Mariner 9. In the initial stages of its retreat the windows in the cap continually changed, but by early 1971 they became fixed and unchanging. The dark features are bare ground, where ice has evaporated from sun-facing slopes. The permanent cap probably contains substantial water, if it is not all water, because a permanent mass of frozen CO_2 would collect water even from Mars' dry atmosphere. The width shown in each photograph is about 100 km.—L. A. Soderblom

(89°N, 200°W; MTVS 4297-47)

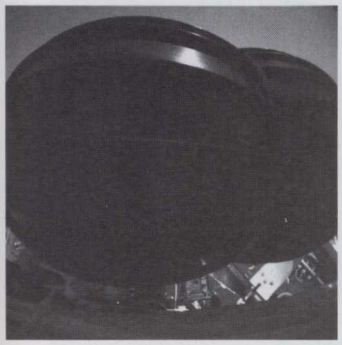
The martian north polar frost cap approached its minimal extent about one-half martian month after summer solstice on October 12, 1972. The cap is about 1000 km across. Its topography and the curved patterns in the interior of the frost cap are interpreted as a series of stacked, slightly concaved plates, the upper one of less areal extent, with edges that have been smoothed and modified. The individual plates may consist of from 20 to 40 separate layers, with an aggregate thickness of perhaps one kilometer. The outline of the residual cap and configuration of the interior markings arise from the frostfree Sun-facing slopes along which layers outcrop. A dark collar of rougher textured terrain surrounds the smoother polar-layered sedimentary complex localized in the central regions of both poles.—L. A. Soderblom



Opportunity is due to a heater on the rover's robotic arm. A switch designed to overrule the heater's thermostatic control has not been working. "In the near term, it's not providing any operational constraints," Welch said.

JPL, a division of the California Institute of Technology in Pasadena, manages the Mars Exploration Rover project for NASA's Office of Space Science, Washington, D.C. Images and additional information about the project are available from JPL at http://marsrovers.jpl.nasa.gov and from Cornell University at http://athena.cornell.edu .

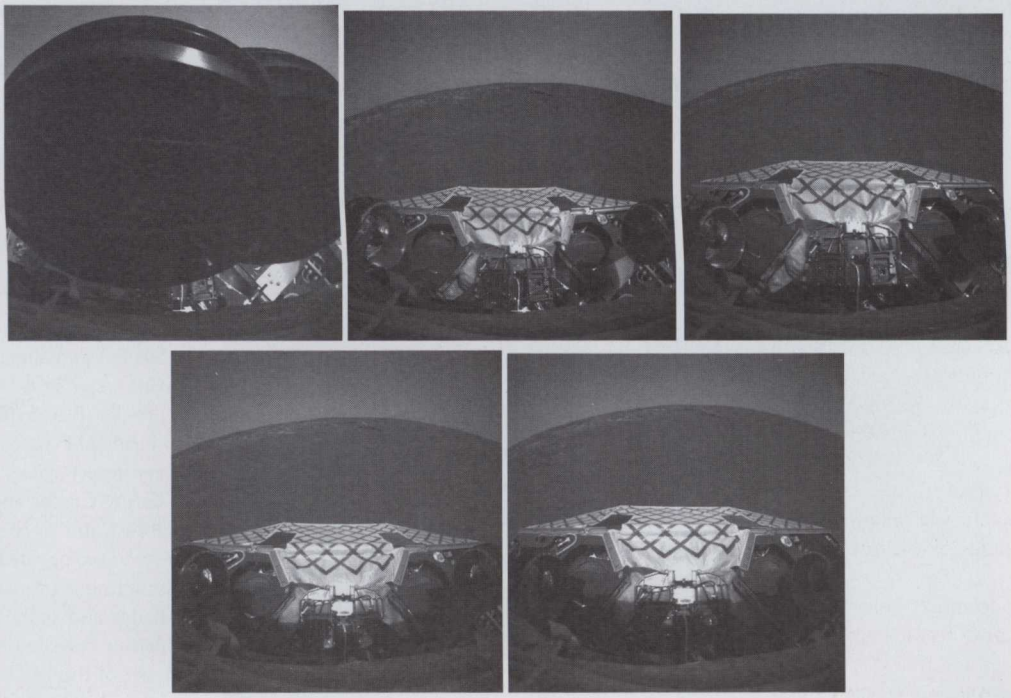
Guy Webster (818) 354-5011 Jet Propulsion Laboratory, Pasadena, California
Donald Savage (202) 358-1547 NASA Headquarters, Washington, D.C.
NEWS RELEASE: 2004-43



An Opportunity to Rise

This image shows the Mars Exploration Rover Opportunity's wheels in their stowed configuration. As of 9:00 a.m. January 28, 2004, the rover had deployed its wheels and completed the first half of the stand-up process. This image was taken at Opportunity's landing site, Meridiani Planum, by the hazard-identification camera.

Image credit: NASA/JPL



A Higher Opportunity (Animation sequence)

This animation strings together images from the rover's front hazard-identification camera taken during the first half of the stand-up process of the Mars Exploration Rover Opportunity at Meridiani Planum, Mars. The first frame shows the rover's wheels tucked under in pre-stand-up position. The following frames show the first two stages of the stand-up process, in which the rover elevates itself and unfolds the wheels.

Image credit: NASA/JPL



Computers and the Cosmos

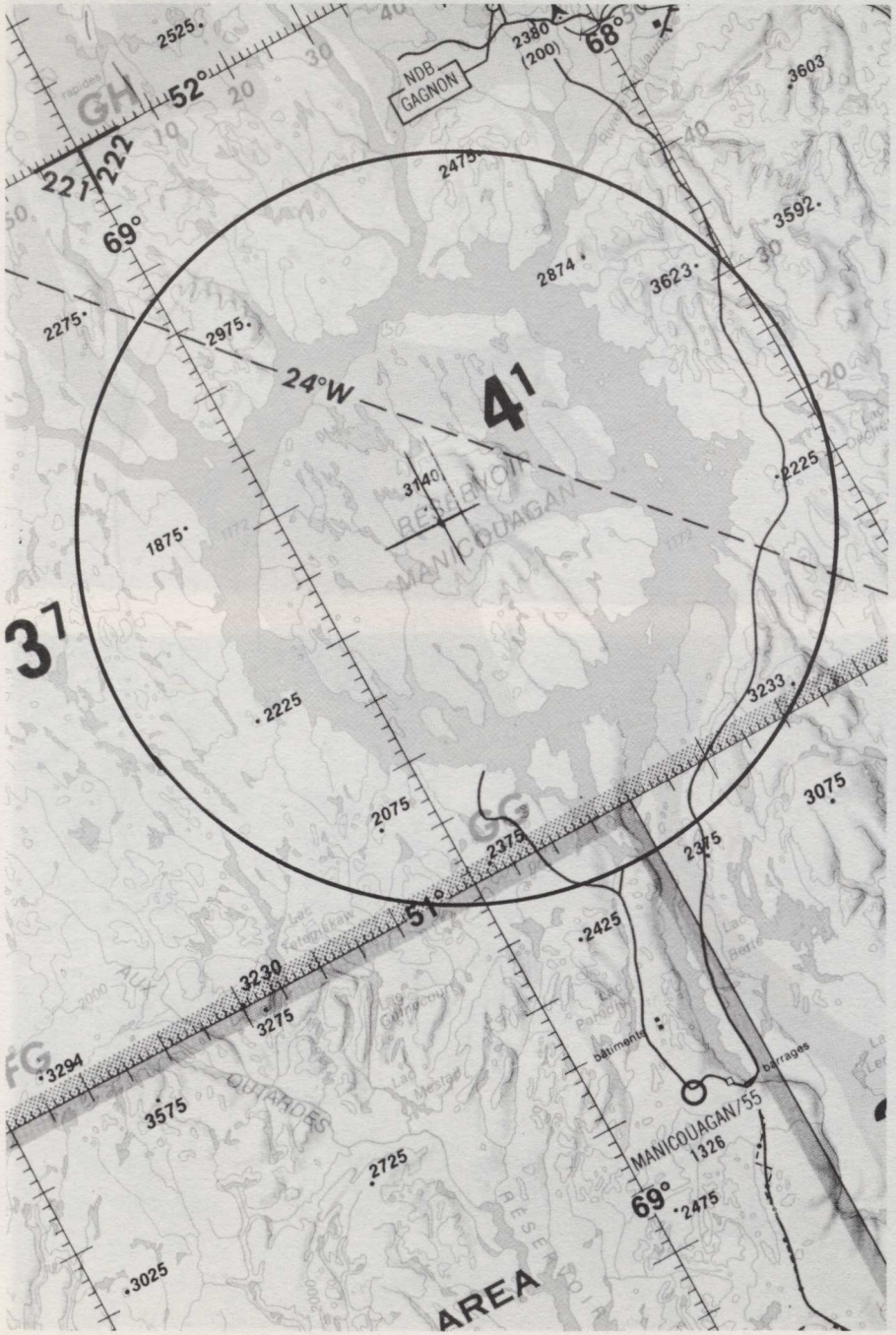
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QUEBEC, CANADA

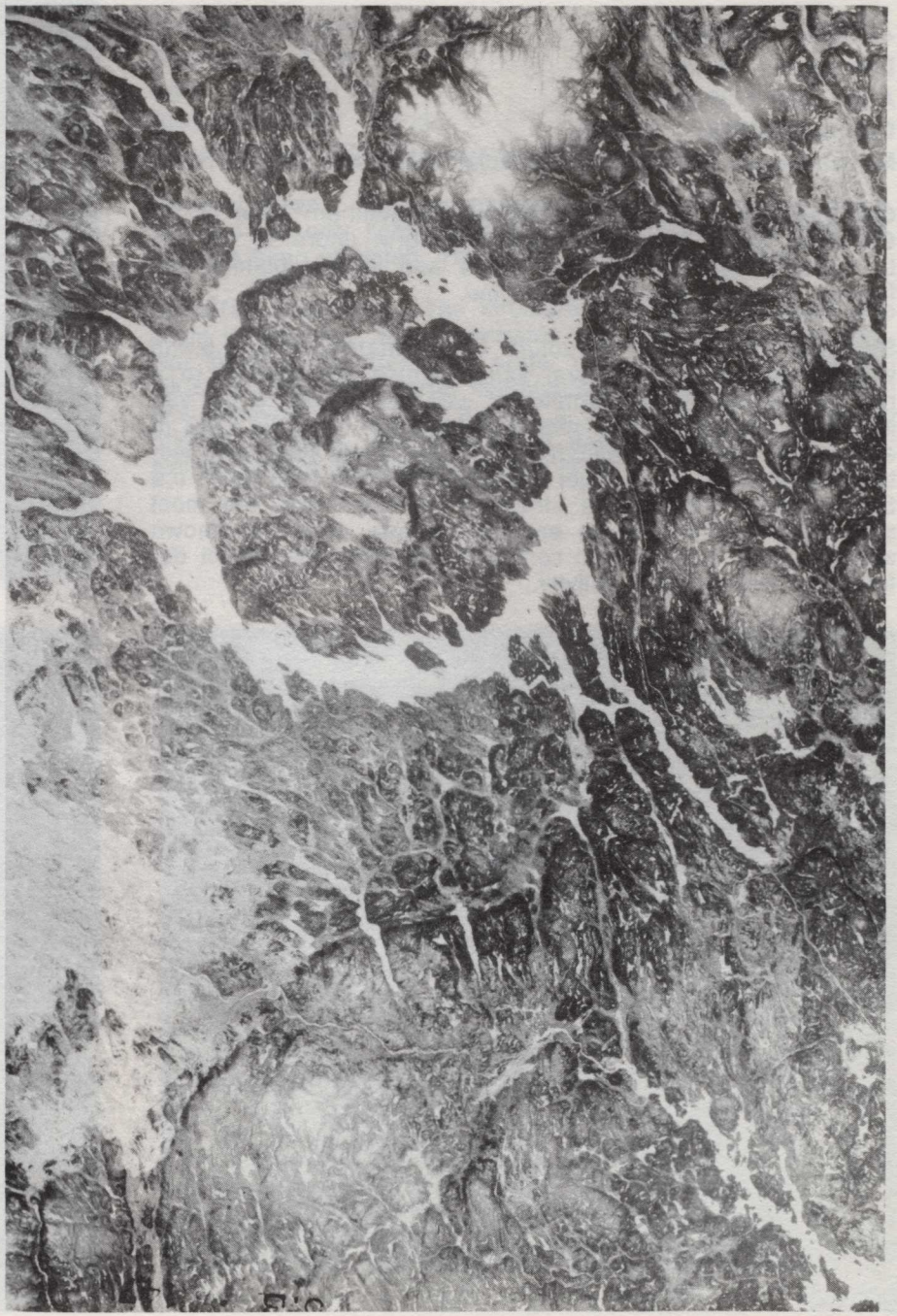
51°23'N; 68°42'W
Diameter: 100 km
Age: 212 ± 2 m.y.
Discernability: Excellent
Morphology: Large annular lake

General Area: Heavily timbered area of the Canadian Shield. The topography is generally rugged and the area has been glaciated, with north-south ice movement. The target rocks are crystalline.

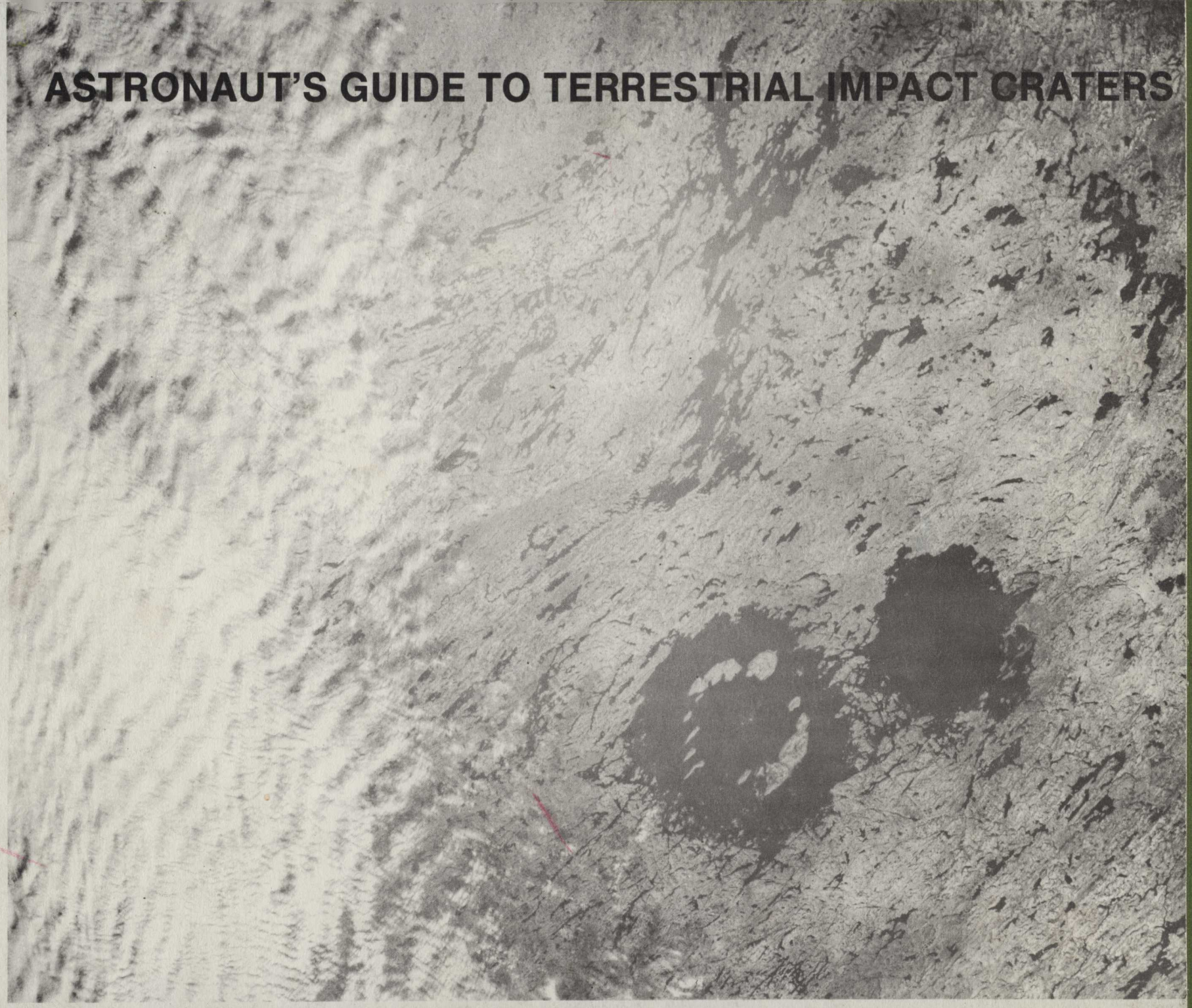
Specific Features: The structure may be divided into a number of morphologic elements. The most striking is the ~70 km diameter annular depression filled by the waters of the Lac Manicouagan reservoir. The annular depression is interpreted as the glacially overdeepend expression of the interior contact between the crater floor and inner blocks of the original rim. In the middle of the lake is a dissected plateau capped by ~200 m of impact melt rocks and a series of uplifted peaks ~5 km north of the center. Of particular interest is a fracture halo, which extends out to ~150 km from the center. First noted on Skylab photography, this halo is best developed in the west and south.



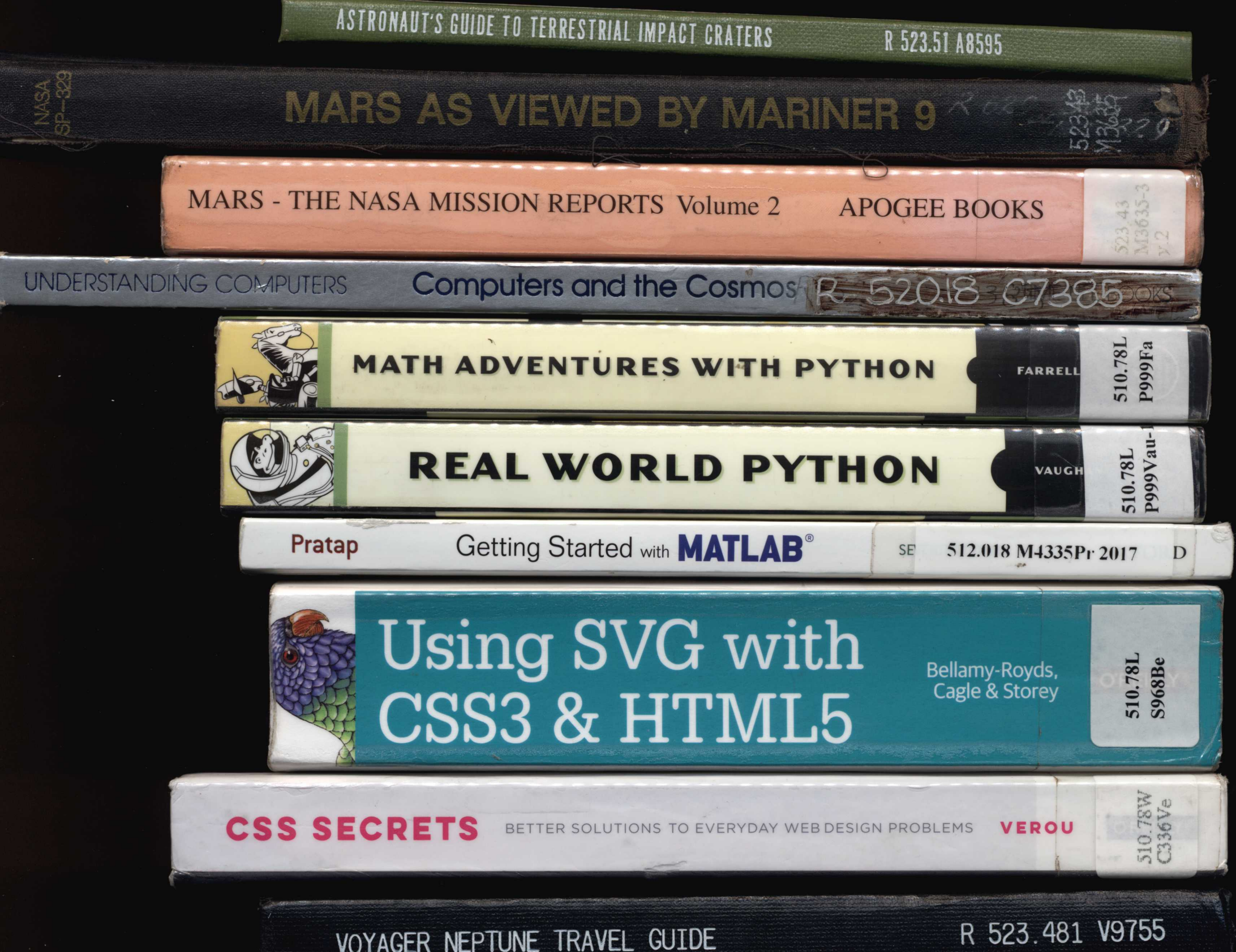

NDB
GAGNON
2380
(200)
68°
52°
GH
2525.
3603
2475.
3592.
69°
2874.
3623.
2275.
2975.
24°W
41
3140
2225.
1875.
37
2225.
3233.
3075.
2075.
GG
2375.
2375.
51°
2425.
3230
3275
FG
3294
3575
OUTARDES
2725.
MANICOUAGAN/55
1326
69°
2475.
3025.
AREA



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